Local Service and Business Intelligence Database

Local Service Insights | Automotive services in Columbia, Missouri
Mayowa, Cailin, Nebiyu, Braiden | CS 284 / CS 430 | Fall 2026

Purpose and scope

Our database helps students and residents find an automotive service provider in Columbia and check the source behind each recorded offering. A business is included only when its name, a physical Columbia street location, and at least one service can be verified. Each distinct business has one selected location. Repeated listings and extra branches do not increase the business count.

Businesses	Services	Categories	Offerings	Source pages
15	4	2	20	20

The sample records 20 offerings across four service types: oil change, brake service, tire rotation, and wheel alignment. The categories are Routine maintenance and Repair and adjustment. This is a limited sample, not a complete list of what each business offers.

Sources and cleaning

The source register contains 20 pages from 18 publisher labels. Provider websites support the service claims. Three address checks compare a provider page with a separate directory: All-Star and AAA (S04/S18), Midas and Michelin (S09/S19), and Walt's and BBB (S13/S20). The checks confirm the listed street locations; they are not on-site inspections. Both URLs, observed values, and any differences are recorded in data/corroboration.csv.

The original capture contains 45 observations dated October 4, 2026: 15 business identity/location records, 20 offerings, three address checks, and seven review or exclusion records. Cleaning follows two rules: standardize names and addresses without changing their meaning, and map explicit service wording to the four controlled service types. For example, “Oil & Filter Change” becomes “Oil change,” and “Brake Repair & Replacement” becomes “Brake service.” Five before-and-after examples are saved in data/cleaning_examples.csv.

A repeated All-Star footer is merged into one business record. Provider-confirmed name changes link Custom Complete to Dobbs and Mutrux to Norris. Two extra branches are excluded. Tom's website gives conflicting alignment information, so that offering is left out. The original wording remains in data/raw/observations.csv; data/cleaning_note.md explains each decision.

Four business rules

1. Each distinct business has one stable ID and one selected physical location. Duplicate listings and additional branches are not separate businesses.

2. Each service belongs to exactly one category. A business can offer several services, and a service can have several providers. Every included business has at least one recorded offering.

3. Names, address/city, service/category names, and source/raw references are required. A website that cannot be verified is stored as SQL NULL.

4. Each business/service pair is stored once. Business identity, location, and offerings must link to supporting source records and saved observations.

Database structure

BusinessService connects Business and Service using the composite primary key (business_id, service_id). Service.category_id links each service to its category. Source supports business identity, location, and offering evidence through foreign keys. Identity and location have separate source references so they can use different supporting pages. Website is the only nullable field. All foreign keys use RESTRICT for updates and deletions; T1 also checks that every included business has an offering.

The complete ER diagram is on page 3. README lists table purposes and keys, and sql/00_schema.sql defines every column. Brand-level service evidence supports the business offering; availability at the selected branch may still need confirmation.

Design reasoning and results

Normalization to 3NF

Start with a combined, deduplicated relation R containing business details, service details, category details, offering evidence, and source metadata for each evidence role. One candidate key is (business_id, service_id), with one selected supporting source for each offering. This relation is different from the raw capture, which can contain several observations of the same fact.

Determinant	Dependent facts
business_id	Name, selected address/city, website, identity/location evidence
service_id	Service name, category_id
category_id	Category name
business_id, service_id	Offering source_id and raw_row_id
source_id (for each role)	Publisher, title, URL, collection date, raw-file path

Business details depend on business_id alone, and service details depend on service_id alone. These are partial dependencies on the combined key. Category and source metadata also introduce transitive dependencies. Splitting R into Business, Service, ServiceCategory, BusinessService, and Source separates those facts. Under the stated dependencies, every nontrivial determinant in each final table is a key, so the tables satisfy 3NF.

This removes an update anomaly: if a business address were repeated on every offering row, one update could leave conflicting addresses. Business stores the address once. Joins reconnect BusinessService to Business and Service, Service to ServiceCategory, and the evidence references to Source. Separate Source aliases reconstruct identity, location, and offering metadata. Foreign keys keep those references valid.

Two findings

Q2: Selecting Columbia and service ID 1 (Oil change) returns 13 distinct recorded providers. Jiffy Lube, for example, is business 10; its oil-change observation is O10, supported by source 10. evidence/Q2.csv includes every provider, its source URL, and raw-row reference. The provider set was checked against the cleaned offering records.

Q3: With a minimum of two providers, Oil change has 13 providers and Brake service has 5. Tire rotation and wheel alignment each have one recorded provider, so they do not pass HAVING >= 2. The manual counts are 13, 5, 1, and 1. Saved outputs and checks are in evidence/Q3.csv and evidence/query_checks.md.

Validation

MySQL 8.0.46-0ubuntu0.24.04.3 executed two clean builds with foreign keys enabled. All five ordered table exports and all six query outputs matched. T1 passed the minimum counts and source checks. T2 rejected a nonexistent business reference (1452), and T3 rejected a duplicate offering pair (1062). T4 tested NOT EXISTS, LEFT JOIN, a missing website, and provider counts in a separate fixture, then removed it.

The execution environment blocked sockets, so validation used MySQL's offline initialization mode with SELECT exports. The submitted four SQL files can also run through a normal MySQL client without the audit harness. The ZIP was extracted and rebuilt successfully. Actual results and logs are in evidence/test_log.md and evidence/package_verification.md.

Limitations and next question

The sample favors businesses with accessible websites and includes only 20 offerings. An offering missing from the database means it was not recorded; it does not prove the business lacks that service. All 15 businesses have verified websites, so Q6 returns no rows. T4 checks missing-website behavior separately. Provider counts describe this sample and cannot establish unmet demand.

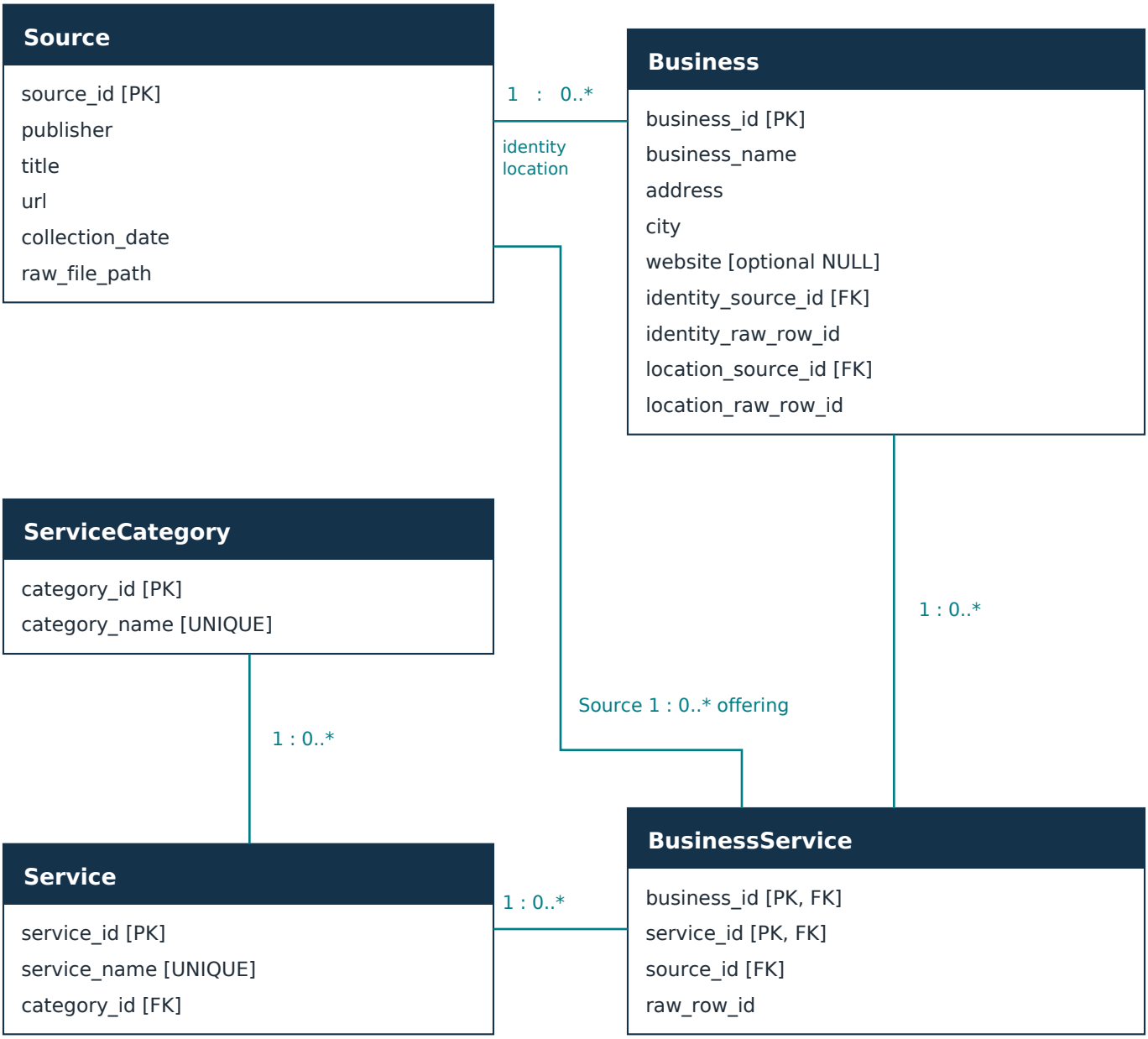
A useful next question is: How would the results change if a complete service inventory were collected for every sampled business?

Team and tools

The instructor has approved the four-member team. Assigned presentation roles are Mayowa: model and keys; Cailin: sources and cleaning; Nebiyu: build and tests; Braiden: results and limitations. Completed reviews and recording contributions will be entered in README. ChatGPT assisted collection, cleaning, SQL, validation, and writing. Facts were checked against original sources, and results against MySQL and saved CSV data. README contains the contribution and tool-use details.

ER diagram and participation

Each child FK requires exactly one existing parent. Parents can exist before child rows load, so structural relationships allow 0..* children. The final sample requires at least one offering per Business, validated by T1. All attributes are required except website. Identity and location are two mandatory Source relationships.



PK / FK participation and composite constraints

BusinessService PK = (business_id, service_id). Business UNIQUE = (name, address, city).